THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A solid hemisphere has radius 21 cm. The total surface area of the hemisphere is: 1

(a) 5544 cm² **(b)** 2772 cm² **(c)** 1386 cm² **(d)** 4158 cm²

2. A right circular cylinder has base radius 7 cm and height 12 cm. The curved surface area of the cylinder is: 1

(a) 836 cm² **(b)** 1848 cm² **(c)** 528 cm² **(d)** 1056 cm²

3. A right circular cylinder has base radius 7 cm and height 3 cm. The volume of the cylinder is: 1

(a) 154 cm³ **(b)** 440 cm³ **(c)** 132 cm³ **(d)** 462 cm³

4. A right circular cone has base radius 21 cm and height 20 cm. The slant height of the cone is: 1

(a) 20.5 cm **(b)** 29 cm **(c)** 19 cm **(d)** 41 cm

5. A right circular cone has base radius 21 cm and height 20 cm. The curved surface area of the cone is: 1

(a) 1914 cm² **(b)** 3300 cm² **(c)** 1320 cm² **(d)** 1386 cm²

6. A solid sphere has radius 28 cm. The surface area of the sphere is: 1

(a) 4928 cm² **(b)** 2464 cm² **(c)** 7392 cm² **(d)** 9856 cm²

7. A solid sphere has radius 31.5 cm. The volume of the sphere is: 1

(a) 12474 cm³ **(b)** 130977 cm³ **(c)** 65488.5 cm³ **(d)** 98232.75 cm³

8. A solid hemisphere has radius 28 cm. The total surface area of the hemisphere is: 1

(a) 2464 cm² **(b)** 7392 cm² **(c)** 4928 cm² **(d)** 9856 cm²

9. A right circular cylinder has base radius 35 cm and height 4 cm. The curved surface area of the cylinder is: 1

(a) 15400 cm² **(b)** 880 cm² **(c)** 8580 cm² **(d)** 1760 cm²

10. A right circular cylinder has base radius 14 cm and height 3 cm. The volume of the cylinder is: 1

(a) 1496 cm³ **(b)** 1848 cm³ **(c)** 264 cm³ **(d)** 616 cm³

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A cuboid measures 15 cm by 10 cm by 6 cm. Find the total surface area of the cuboid. 2

12. A right circular cylinder has base radius 35 cm and height 2 cm. Find the total surface area of the cylinder. 2

13. A solid hemisphere has radius 42 cm. Find the volume of the hemisphere. 2

14. A cuboid measures 20 cm by 9 cm by 5 cm. Find the total surface area of the cuboid. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A right circular cone has base radius 21 cm and height 20 cm. Find the volume of the cone. 3

16. A right circular cone has base radius 21 cm and height 20 cm. Find the total surface area of the cone. 3

17. The total surface area of a cube is 294 cm^2. Find the volume of the cube. 3

OR

A right circular cone has base radius 35 cm and height 12 cm. Find the volume of the cone. 3

18. A right circular cone has base radius 35 cm and height 12 cm. Find the total surface area of the cone. 3

OR

The total surface area of a cube is 384 cm^2. Find the volume of the cube. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A solid metal cuboid 24 cm by 24 cm by 6 cm is melted and recast into cubes of edge 3 cm. Find the number of cubes obtained. 5

OR

A solid metallic sphere of radius 8 cm is melted and recast into small solid spheres of radius 2 cm. Find the number of small spheres obtained. 5

20. A solid metal cuboid 24 cm by 24 cm by 18 cm is melted and recast into cubes of edge 6 cm. Find the number of cubes obtained. 5

OR

A solid metallic sphere of radius 15 cm is melted and recast into small solid spheres of radius 5 cm. Find the number of small spheres obtained. 5

END OF QUESTION PAPER — Total: 40 Marks

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